PUMUDU RAJAKARUNA

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SUMMARY

Computer Engineering graduate from the University of Peradeniya with two years of continuous industry experience in mobile and backend development. Research experience in machine learning and data science, with a published research paper and participation in academic research conferences. Strong interest in teaching, research, and academic activities.

EDUCATION

Bachelor of the Science of Engineering (Honours) specializing in Computer Engineering,
University of Peradeniya, Sri Lanka 2021 – Aug 2025
GPA: 3.45 / 4.0

St. Anne's College, Kurunegala 2010 – 2019
Secondary education

PUBLICATIONS

P. Rajakaruna et al., "Ensemble Machine Learning Based Clinical Decision Support System for Optimizing Hemodialysis Parameters in End-Stage Renal Disease Patients," 2026 IEEE International Research Conference on Smart Computing and Systems Engineering (SCSE), Kelaniya, Gampaha, Sri Lanka, 2026, pp. 1-6, doi: 10.1109/SCSE70081.2026.11499901.

- First author.
- Conducted in collaboration with the Faculty of Engineering, Faculty of Science, Faculty of Medicine, University of Peradeniya, and the Teaching Hospital Peradeniya.

AWARDS

Best Oral Presentation Award - Clinical and Biomedical Stream, National Health Research Symposium organized by the Education, Training and Research Unit, Ministry of Health and Mass Media - Nov 2025

Best Paper Award - Health Analytics Track, 9th IEEE International Research Conference on Smart Computing and Systems Engineering - 2026

EXPERIENCE

Software Engineer Dec 2025 - Oct 2026
KAST - A stablecoin neobank | Hybrid

- Worked primarily as a backend developer, contributing to scalable API development for payment and financial services.
- Integrated third-party financial services, including stablecoin payouts, USD/EUR fiat deposits, and Brazilian BRL pay-ins and payouts, collaborating directly with external provider teams.
- Contributed to Functional Specification Documents (FSDs), production debugging using AWS CloudWatch and Datadog, and coordination with cross-functional teams, including frontend, design, QA, and external partners.

Technologies and languages used: Golang, Gin, PostgreSQL, Bruno, AWS, Jira, Notion, Docker

Associate Software Engineer Jan - Dec 2025
Centre for Technology and Innovation, LB Finance PLC | Remote | Part-time

- Final phase contributor to LB CIM mobile banking application revamp (mobile & web), now live.
- Built core features: redesigned dashboard and savings account flow.
- Developing new features incl. GovPay integration and personal loan creation.

Software Engineer - Intern Jul 2024 – Jan 2025
Centre for Technology and Innovation, LB Finance PLC | Onsite.

- Delivered core features: Fixed Deposit creation and Lease/Loan requests (live in production).
- Contributed to bug fixing across most app modules.
- Supported LB eConnect (web portal) with feature upgrades and backend revamp of Fixed Deposit flow.

Technologies and languages used: Java, Spring Boot, MySQL, React.js, Dart, Flutter, OpenAPI, Android Studio

Casual Instructor(Teaching Assistant) in the Department of Computer Engineering

CO253 - Introduction to Programming and Networking for Electrical Engineering Feb – June 2024
- Conducted labs in C programming, networking incl. Cisco Packet Tracer & Wireshark.

CO223 - Computer Communication Networks Oct 2023 – Jan 2024
- Delivered networking labs incl. Cisco Packet Tracer & Wireshark and supported student learning.

PROJECTS

AI-driven Renal Care Management – Teaching Hospital Peradeniya / Final Year Project/ Group Feb 2025 – Present
React | Python | Django | MongoDB | Base Web

- Conducted a systematic review and collected hospital-documented data to guide model development.
- Performed exploratory data analysis (EDA), trained ML models to predict key dialysis parameters, and contributed to frontend development and ML-backend integration.
- Collaborated with healthcare professionals to gather system requirements and validate clinical insights for the renal care system.

Smart Chess Board / Group Nov 2023 – Mar 2024
Flutter | Node.js | AWS Elastic Beanstalk

- Developed a platform connecting a physical chess board to a mobile app for remote gameplay.
- Contributed to UI design, backend development, and cloud deployment.
- Authored the product design manual.

EXTRA-CURRICULAR

Hackers' Club, University of Peradeniya Feb 2025
Prepared and delivered a hands-on mobile development workshop for undergraduates as part of the Developer Series, including developing tutorial materials and guiding participants through practical exercises.

AIESEC in Kandy 2021 / 2022

REFERENCES

Dr. Upul Jayasinghe
Senior Lecturer - Department of Computer Engineering, Faculty of Engineering, University of Peradeniya
Email: upuljm@eng.pdn.ac.lk